Football Data Engineering Pipeline

End-to-End Big Data Analysis & Visualization using PySpark, Databricks, and Power BI (2000 - 2015 Statistics)

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Executive Project Overview

The Challenge

Processing over 15 years of complex football match data to extract actionable insights. The objective was to move from **unstructured CSV logs** to a highly optimized **Gold-standard Analytics Layer**.

Key Objectives

- Distributed Data Processing with PySpark
- Medallion Architecture implementation
- Cloud-Native Deployment on Databricks

Medallion Data Architecture

Bronze Layer

Raw Ingestion: Capturing historical football CSV data with schema inference and initial sanitization.

Silver Layer

Refined Processing: Feature engineering, win/loss/tie indicators, and large-scale joins across seasons.

Gold Layer

Business Logic: Advanced KPIs, Window Ranking functions, and optimized Parquet partitioning.

Bronze: Efficient Raw Ingestion

Distributed Reading

Using **Spark's distributed engine** to read large-scale CSV logs. Optimized by selecting only high-value dimensions like Match ID, Division, and Team Metrics.

- Schema Inference enabled
- Predicate Pushdown at the source
- SparkSession configuration for optimal RAM

Silver: Transforming Indicators

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Seasons Processed (2000-2015)

Transformation Logic

Conversion of categorical match results (H/A/D) into numerical binary indicators for scalable aggregation.

Complex Joins: Merging 'Home' and 'Away' DataFrames to create a 360-degree performance view for every European club.

Gold: Ranking & Window Functions

- **Spark Window Functions:** Implemented `partitionBy("Season")` to rank teams without disrupting the distributed dataset structure.
- **Multi-Factor Sorting:** Ranking champions based on both Win Percentage and Goal Differentials.
- **Data Granularity:** Moving from match-level logs to season-level executive summaries.

Performance Metrics Dashboard

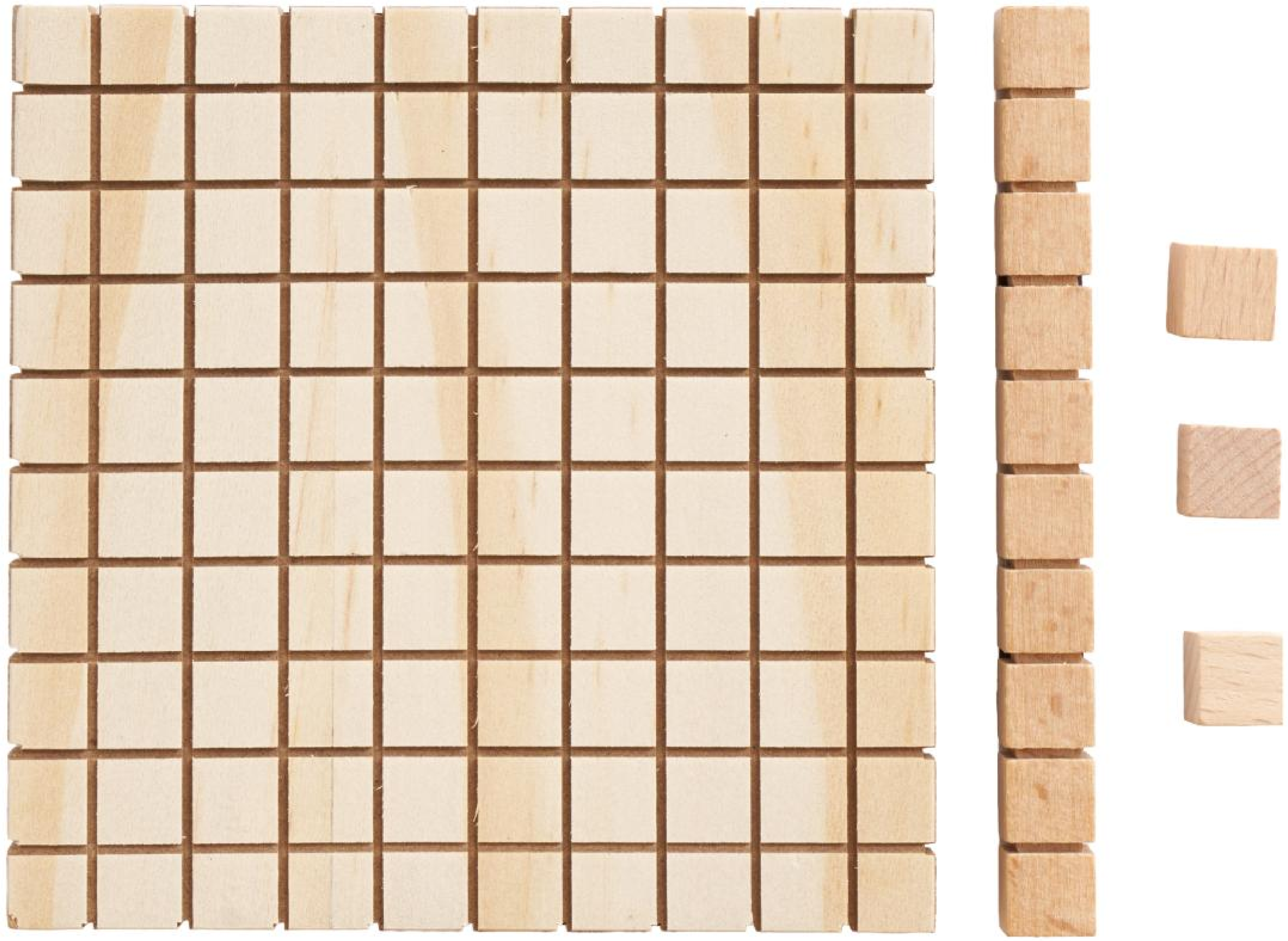
Calculated KPIs: Percentage of victories, Goal Differentials per Season, and Defensive Stability ratios.

Storage Optimization

Parquet vs. CSV

Implemented **Columnar Storage** using Parquet, resulting in up to 80% reduction in storage footprint and faster Power BI querying.

Partitioning Strategy: Partitioned Gold layer data by Season to enable "Data Skipping" during analysis.



Power BI: Impactful Insights

Connecting optimized Parquet data to a dynamic presentation layer. Features include interactive slicers for Season/Team comparison and Real-Time KPI Cards.

Problem Solving & Debugging

DBFS Access

Overcame `DBFS_DISABLED` security restrictions by pivoting to Workspace-local protocols for file ingestion.

Precision Logic

Fixed 6000% overflow errors in Power BI by implementing DAX Measures for proper percentage normalization.

Conclusion & Successes

- **Scalability:** Pipeline ready for billions of rows of historical match data.
- **Efficiency:** Transition from raw CSV to partitioned Parquet optimized costs.
- **Cloud-Ready:** Seamless migration from Google Colab to Databricks Ecosystem.

Questions & Answers

Thank you for your attention. I am now open to your questions regarding the architecture, logic, or implementation.

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Image Sources

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